

```
> Packages > S319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > D668427361B00E7E05AF05484488873738
1  def is_valid_username(username):
2      if len(username) < 5 or len(username) > 15:
3          return False
4      if not username[0].isalpha():
5          return False
6      for char in username:
7          if not (char.isalnum() or char == '_'):
8              return False
9      return True
10 #test cases for the is_valid_username function
11 assert is_valid_username("User123") == True
12 assert is_valid_username("12User") == False
13 assert is_valid_username("Us er") == False
14 print("All test cases for is_valid_username passed!")
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

```
PS C:\Users\Sushmitha> & C:\Users\Sushmitha\AppData\Local\Microsoft\WindowsApps\python3.10.0\python.exe C:\Users\Sushmitha\AppData\Local\Microsoft\WindowsApps\python3.10.0\python.exe tsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/D668427361B00E7E05AF05484488873738
All test cases for is_valid_username passed!
PS C:\Users\Sushmitha>
```

Task Description #2 (Even–Odd & Type Classification – Apply AI for Robust Input Handling)

- Task: Use AI to generate at least 3 assert test cases for a function `classify_value(x)` and implement it using conditional logic and loops.
- Requirements:
 - o If input is an integer, classify as "Even" or "Odd".
 - o If input is 0, return "Zero".
 - o If input is non-numeric, return "Invalid Input".

Example Assert Test Cases:

```
assert classify_value(8) == "Even"
```

```
assert classify_value(7) == "Odd"
```

```
assert classify_value("abc") == "Invalid Input"
```

Expected Output #2:

- Function correctly classifying values and passing all test cases.

```
2303A51666_LabAssignment-8.py ×  
ickages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > D668427361B00E7E05AF054B4488873738B6B50C > transfers > 2026-07 > 2303A51666_LabAssignment-8.py > ...  
  
15  
16 def classify_value(x):  
17     if x < 0:  
18         return "Negative"  
19     elif x == 0:  
20         return "Zero"  
21     elif x%2 == 0:  
22         return "Even"  
23     else:  
24         return "Odd"  
25 # Test cases for the classify_value function  
26 assert classify_value(8) == "Even"  
27 assert classify_value(7) == "Odd"  
28 assert classify_value("abc") == "Invalid Input"  
29  
  
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS  
PS C:\Users\Sushmitha> & C:\Users\Sushmitha\AppData\Local\Microsoft\WindowsApps\python3.13.exe c:/Users/Sushmitha/AppData/Local/Packages/5319275A.Wha  
tsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/D668427361B00E7E05AF054B4488873738B6B50C/transfers/2026-07/2303A51666_LabAssignment-8.py  
Traceback (most recent call last):  
File "c:\Users\Sushmitha\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\D668427361B00E7E05AF054B4488873738B6B50C  
\transfers\2026-07\2303A51666_LabAssignment-8.py", line 28, in <module>  
    assert classify_value("abc") == "Invalid Input"  
           ~~~~~~  
File "c:\Users\Sushmitha\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\D668427361B00E7E05AF054B4488873738B6B50C  
\transfers\2026-07\2303A51666_LabAssignment-8.py", line 17, in classify_value  
    if x < 0:  
       ~~~~  
TypeError: '<' not supported between instances of 'str' and 'int'  
PS C:\Users\Sushmitha>
```

Task Description #3 (Palindrome Checker – Apply AI for String Normalization)

- Task: Use AI to generate at least 3 assert test cases for a function is_palindrome(text) and implement the function.

- Requirements:
 - o Ignore case, spaces, and punctuation.
 - o Handle edge cases such as empty strings and single characters.

Example Assert Test Cases:

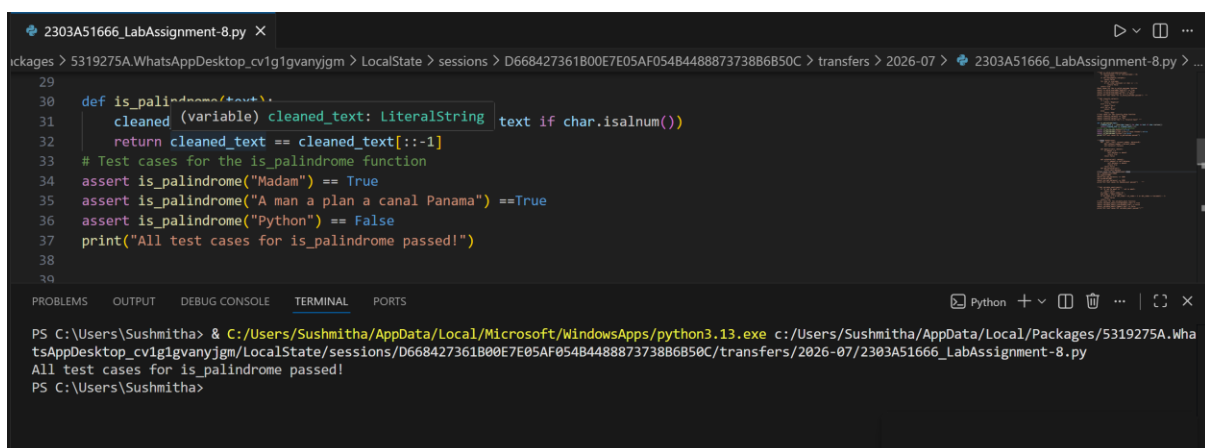
```
assert is_palindrome("Madam") == True
```

```
assert is_palindrome("A man a plan a canal Panama") ==  
True
```

```
assert is_palindrome("Python") == False
```

Expected Output #3:

- Function correctly identifying palindromes and passing all AI-generated tests.



The screenshot shows a VS Code editor with a file named '2303A51666_LabAssignment-8.py'. The code defines a function `is_palindrome(text)` that cleans the text (removes non-alphanumeric characters and spaces) and checks if it is a palindrome. Below the function, three test cases are written using `assert` statements: `assert is_palindrome("Madam") == True`, `assert is_palindrome("A man a plan a canal Panama") == True`, and `assert is_palindrome("Python") == False`. A `print` statement at the end says "All test cases for is_palindrome passed!". The terminal at the bottom shows the command to run the script and the output: "All test cases for is_palindrome passed!".

```
29  
30 def is_palindrome(text):  
31     cleaned_text = LiteralString(text if char.isalnum())  
32     return cleaned_text == cleaned_text[::-1]  
33 # Test cases for the is_palindrome function  
34 assert is_palindrome("Madam") == True  
35 assert is_palindrome("A man a plan a canal Panama") == True  
36 assert is_palindrome("Python") == False  
37 print("All test cases for is_palindrome passed!")  
38  
39
```

```
PS C:\Users\Sushmitha> & C:/Users/Sushmitha/AppData/Local/Microsoft/WindowsApps/python3.13.exe c:/Users/Sushmitha/AppData/Local/Packages/5319275A.WhatsAppDesktop_cv1g1gvanyjgm/LocalState/sessions/D668427361B00E7E05AF0548448887373886B50C/transfers/2026-07/2303A51666_LabAssignment-8.py  
All test cases for is_palindrome passed!  
PS C:\Users\Sushmitha>
```

Task Description #4 (BankAccount Class – Apply AI for Object-Oriented Test-Driven Development)

- Task: Ask AI to generate at least 3 assert-based test cases for a BankAccount class and then implement the class.
- Methods:
 - o deposit(amount)
 - o withdraw(amount)
 - o get_balance()

Example Assert Test Cases:

```
acc = BankAccount(1000)
```

```
acc.deposit(500)
```

```
assert acc.get_balance() == 1200
```

- Fully functional class that passes all AI-generated assertions.

```
> Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > D668427361B00E7E05AF054B4488873738B6B50C > transfers > 2026-07 > 2303A51666_LabAssignment-8-p
1 def is_valid_username(username):
2     if len(username) < 5 or len(username) > 15:
3         return False
4     if not username[0].isalpha():
5         return False
6     for char in username:
7         if not (char.isalnum() or char == '_'):
8             return False
9     return True
10 #test cases for the is_valid_username function
11 assert is_valid_username("User123") == True
12 assert is_valid_username("12User") == False
13 assert is_valid_username("Us er") == False
14 print("All test cases for is_valid_username passed!")
```

- Task: Use AI to generate at least 3 assert test cases for a function `validate_email(email)` and implement the function.

- Requirements:
 - o Must contain @ and .
 - o Must not start or end with special characters.
 - o Should handle invalid formats gracefully.

```
assert validate_email("@gmail.com") == False
```

- Email validation function passing all AI-generated test cases and handling edge cases correctly.

```
> Packages > 5319275A.WhatsAppDesktop_cv1g1gvanyjgm > LocalState > sessions > D668427361B00E7E05AF054B4488873738B6B50C > transfers > 2026-07 > 2303A51666_LabAssignment-8.py X
```

```
1 def validate_email(email):
2     if '@' not in email or '.' not in email:
3         return False
4     at_index = email.index('@')
5     dot_index = email.index('.')
6     if at_index < 1 or dot_index < at_index + 2 or dot_index >= len(email) - 1:
7         return False
8     return True
9
10 # Test cases for the validate_email function
11 assert validate_email("user@example.com") == True
12 assert validate_email("userexample.com") == False
13 assert validate_email("@gmail.com") == False
14 print("All test cases for validate_email passed!")
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

```
PS C:\Users\Sushmitha> & C:\Users\Sushmitha\AppData\Local\Microsoft\WindowsApps\python3.13.exe c:\Users\Sushmitha\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\D668427361B00E7E05AF054B4488873738B6B50C\transfers\2026-07\2303A51666_LabAssignment-8.py
All test cases for validate_email passed!
PS C:\Users\Sushmitha>
```